

Pixel Art Is a Palette, Not a Downscale — Supplementary Material

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001 1. What the body left out

002 The body reports what a reader needs to use the method.
003 This document carries the rest: the data pipeline in enough
004 detail to rebuild the corpus, the results that did not earn a
005 place in the recipe, and the measurements behind claims the
006 body states in one sentence. Section numbers here are inde-
007 pendent of the body.

008 2. Corpus construction

009 **From sheets to sprites.** Source material arrives as sprite
010 sheets and loose PNGs. Whole images whose longest side
011 is at most 64 px are kept as they are — these are the na-
012 tively tiny sprites the low-resolution buckets are starving
013 for. Larger images are treated as sheets and cut into alpha-
014 connected components between 8 and 64 px. Every cut
015 records the archive, the member file and the licence it came
016 with.

017 **Filters.** A candidate is discarded unless its shorter side is
018 at least 10 px, its aspect ratio is at most 3:1, the fraction
019 of pixels with $\alpha > 16$ lies in $[0.15, 0.97]$, it has at least
020 four distinct colours after quantising each channel to 8 lev-
021 els, and the standard deviation of its opaque RGB is at least
022 18. These reject background patches, single-colour blocks,
023 and the near-empty crops that connected-component cutting
024 produces at sheet margins.

025 **Deduplication.** Every surviving sprite is hashed by the
026 exact bytes of its RGBA array together with its shape. Within
027 the original corpus this finds 4,888 extra copies. Because
028 hold-out sets are defined by path, every copy of a held-
029 out sprite is also excluded from training, not only the copy
030 that was drawn. Near-duplicates — animation frames of the
031 same character — are not removed from training: an abla-
032 tion that collapsed them by an 8×8 luminance-plus-alpha
033 average hash lost more data than it gained diversity (training
034 rows 64,990 to 49,984) and scored worse at equal steps, so
035 only the per-source cap below is applied.

Bucket policy. A sprite may feed a resolution bucket only
if its native size is at most $1.5\times$ that bucket. This is the
change that stops 25–64 px artwork from becoming a 16 px
target. It costs training rows: in the enlarged corpus, the
one whose training log we still have, it refuses 154,312 of
272,690 candidate (sprite, bucket) pairs. It is the single
largest contributor to the data-side gain.

Added material. 14,999 sprites cut from three permis-
sively licensed OpenGameArt collections (CC-BY-4.0, CC-
BY-SA-3.0, OGA-BY-3.0) pass the filters and cross-corpus
deduplication. A second pass drops sheets whose member
name matches a font, UI or text pattern (367 sprites), col-
lapses near-identical animation frames within a sheet, and
caps any single source sheet at 60 sprites so one anima-
tion cannot dominate a bucket. The survivors have a median
palette of eight colours and 6,028 of them are natively 24 px
or smaller.

Captions. Sprites are captioned by a vision-language
model shown numbered grids of 20 sprites on a checker-
board background, which is an order of magnitude cheaper
than one call per sprite. A text-only classifier then labels
each caption object / effect / fragment / glyph / junk; on
the 2,980 sprites that also carry image-based labels the two
agree 82.5% of the time (90.5% recall on objects), which is
enough to use the labels for an ablation and not enough to
filter the training corpus with them.

Licences. The sprites added in this work carry per-file
records of archive, member and licence. The older mate-
rial predates that bookkeeping; we release the manifests for
what we added and the scripts that rebuild everything else
from public archives.

- 067 **3. Data interventions that did not earn a place**
- 068 **4. Cross-resolution self-guidance: the negative**
- 069 **result in full**
- 070 **5. Ablations on the model rather than the de-**
- 071 **coder**
- 072 **6. Resolutions outside the target range**
- 073 **7. Alternative reference sets**
- 074 **8. Prompt alignment in full**
- 075 **9. Licences and release**